

The record costs disturbance: a floor for every instrument

For every protocol of marks of any kind (any unitary coupling of the body to an apparatus, any pointer, noise that depends on the body or not, any apparatus state), deciding at error probability ϵ between free motion and a constant force F over a duration τ , whatever the body's initial state under each hypothesis, requires

$$\frac{s}{8} \sum_j \Delta(\hat{D}_j) + \frac{J}{2} \sum_j \Delta(\hat{X}_j) \geq \hbar \arcsin(1 - 2\epsilon), \quad s = \frac{F\tau^2}{2m}, \quad J = F\tau,$$

where $\hat{D}_j$ and $\hat{X}_j$ are the impulse and the position jump that mark j gives the body, as Heisenberg operators on body and apparatus, and Δ is their spread, the supremum over the states that enter the mark in the interpolating processes of the proof; for disturbances that are apparatus operators these are the apparatus's own states. The reading error does not appear. The initial state may differ between the two hypotheses, which is what an unknown initial state means for a test between them, and the constants $1/8$ and $1/2$ are the price of that: a test allowed to assume the same initial state under both faces the coefficients s and J instead. The sagitta of Lemma X is paired with the momentum disturbance and the impulse of Proposition I with the position disturbance, the same pairing as in the aperture theorem of the probabilistic note, $JL + sP \geq \hbar \arcsin(1 - 2\epsilon)$. The constant $\arcsin(1 - 2\epsilon)$ comes from running the proof with the Bures angle in place of total variation (Theorem P of the path-length note, 2026-09-23); the proof as written gives the weaker $1 - 2\epsilon$, and the path-length note also joins this bound and the aperture bound in one accounting.

This is option 2 of the plan set on 2026-09-22 for extending Theorem A, and it closes that plan. Ozawa's error-disturbance relation (PRA **67**, 042105, 2003, metadata), which was the expected tool for body-dependent noise, turns out to be unnecessary, though $\hat{D}_j$ and $\hat{X}_j$ are his disturbance operators. The ingredients are standard: Ozawa's disturbance operators, Mandelstam-Tamm along a path, and a symmetry-based hybrid argument, the reasoning behind information-disturbance bounds for covariant families and behind Wigner-Araki-Yanase-type bounds (Marvian and Spekkens, Nat. Commun. **5**, 3821, 2014; Tajima, Shiraishi and Saito, PRL **121**, 110403, 2018; Kuramochi and Tajima, PRL **131**, 210201, 2023; metadata). What is claimed here is the combination and its reading in Newton's quantities. The floor is a statement about disturbance alone, because the offset between the two hypotheses has to be carried through every mark, and what a mark does to that offset is fixed by what it does to the body. For von Neumann marks, which disturb only momentum, the theorem is the recoil theorem of the recoil note; for marks that read momentum, which disturb position, it is new. Exploratory; no ledger promotion.

1. Setting

Transverse coordinate $\hat{y}$, momentum $\hat{p}$, mass m , on $[0, \tau]$. Mark j acts impulsively at t_j by a unitary U_j on body $\otimes$ apparatus j , after which a pointer of apparatus j is read; the apparatus starts in any state, and couplings may depend on earlier readings. Define the mark's disturbances

$$\hat{D}_j = U_j^\dagger \hat{p} U_j - \hat{p}, \quad \hat{X}_j = U_j^\dagger \hat{y} U_j - \hat{y},$$

operators on body and apparatus, with no restriction on how they depend on either. Under F the body feels a force whose displacement relative to free motion from rest is $P(t)$, $P(0) = P'(0) = 0$. A test decides at error ϵ when its equal-prior error is at most ϵ for every pair of initial states.

For a line $a + bt$ set $c(t) = P(t) - a - bt$ and $w(t) = (c(t), mc'(t))$, the **offset**: if the initial state under F is the initial state under I translated by $w(0) = (-a, -mb)$, then between marks

the state under F is the state under I translated by $w(t)$, up to a phase, because free evolution carries a translation by (c, mc') to one by $(c + c'\delta t, mc')$ and the force adds $(P(t') - P(t) - P'(t)\delta t, m(P'(t') - P'(t)))$. Write $T(w) = e^{-iG_w}$, $G_w = (c\hat{p} - mc'\hat{y})/\hbar$, for the translation by w .

2. The theorem

Theorem U. For every line $a + bt$ and every initial state σ ,

$$1 - 2\epsilon \leq \frac{1}{\hbar} \sum_j \sup \Delta(c_j \hat{D}_j - mc'_j \hat{X}_j),$$

with $c_j = c(t_j)$, $c'_j = c'(t_j)$, and the supremum over the conditional states that can enter mark j when the protocol is run from σ with the offset carried through the earlier marks and then removed, translated by $sw(t_j)$ for $s \in [0, 1]$. For marks whose disturbances are apparatus operators, these are the apparatus's own states and the body drops out.

Proof. Take the pair of initial states σ under I and $T(w(0))\sigma T(w(0))^\dagger$ under F; a test deciding at error ϵ has $\text{TV} \geq 1 - 2\epsilon$ between their record distributions. Let hybrid H_r run F with the offset through mark r , remove the offset immediately after mark r , and run I afterwards. By §1, removing the offset at any time between marks gives the same process, so H_0 is I from σ , H_k is F from the translated state, and H_{r-1} and H_r agree up to mark r and differ only there: with ρ the state entering mark r in H_{r-1} and $w = w(t_r)$, H_{r-1} applies U_r and H_r applies $T(w)^\dagger U_r T(w) = E U_r$ with $E = T(w)^\dagger U_r T(w) U_r^\dagger$. The remainders are the same channel, so by data processing and the triangle inequality

$$\text{TV} \leq \sum_r \frac{1}{2} \|E\rho' E^\dagger - \rho'\|_1, \quad \rho' = U_r \rho U_r^\dagger.$$

For $E(u) = e^{iuG} U_r e^{-iuG} U_r^\dagger$, $u \in [0, 1]$, $G = G_w$, one has $\frac{d}{du} E(u) = iK(u)E(u)$ with $K(u) = e^{iuG}(G - U_r G U_r^\dagger)e^{-iuG}$. The generators are unbounded; the step below assumes that $E(u)$ maps the purification into the domain of $K(u)$ with $u \mapsto E(u)|\Psi\rangle$ differentiable, which holds on a dense set of states and extends to the rest by continuity, and it allows $\Delta K = \infty$, in which case the bound is vacuous (a sharp position mark pays that). The Mandelstam–Tamm bound in the geometric form of Anandan and Aharonov bounds the Fubini–Study angle between a purification of ρ' and its image under $E = E(1)$ by $\int_0^1 \Delta K(u) du$, each spread taken in the state $E(u)\rho' E(u)^\dagger$, and the trace distance by the sine of that angle, as in Theorem 2 of the probabilistic note. Conjugating back, the spread of $K(u)$ in $E(u)\rho' E(u)^\dagger$ equals the spread of $U_r^\dagger G U_r - G$ in $T(uw)\rho T(uw)^\dagger$, and

$$U_r^\dagger G_w U_r - G_w = \frac{1}{\hbar} (c \hat{D}_r - mc' \hat{X}_r).$$

Adaptive couplings replace each term by its supremum over the conditional states, as in the hybrid argument of the probabilistic note. $\square$

Corollary U1 (constant force). Take the line of best uniform approximation, $a + bt = \frac{F}{2m}(\tau t - \tau^2/8)$, a valid choice that is optimal for the momentum term alone; with position disturbances present, the minimum of the combined sum over lines can be smaller. Then $|c_j| \leq s/8$ and $m|c'_j| = F|t_j - \tau/2| \leq J/2$, and the spread is a seminorm, so

$$(1 - 2\epsilon)\hbar \leq \frac{s}{8} \sum_j \sup \Delta(\hat{D}_j) + \frac{J}{2} \sum_j \sup \Delta(\hat{X}_j). \quad \square$$

For a general force the same step gives any line's $\max |c|$ and $m \max |c'|$ in place of $s/8$ and $J/2$.

3. What the theorem covers

The marks are impulsive. A coupling that lasts a finite time is covered by slicing it into short impulsive pieces, each charged with the offset at its own time.

Von Neumann marks. $U = e^{-i\lambda\hat{y}\hat{P}_A/\hbar}$ gives $\hat{D} = -\lambda\hat{P}_A$ and $\hat{X} = 0$, so Theorem U is Theorem R of the recoil note, $1 - 2\epsilon \leq \hbar^{-1} \min_{a,b} \sum_j \Delta_j |c_j|$.

Marks that read momentum. $U = e^{-i\mu\hat{p}\hat{Q}_A/\hbar}$ shifts the pointer momentum by $-\mu\hat{p}$ and gives $\hat{D} = 0$, $\hat{X} = \mu\hat{Q}_A$. A momentum mark therefore pays in position jumps, and its term is paired with the impulse J : recording that the body has gained momentum costs an undetermined displacement.

Marks with body-independent noise that also displace position. The additive-noise note shows that Theorem R holds for them with the recoil spread alone, since the jump $\hat{X}_j$ then commutes with the reading and the kick. Theorem U gives the weaker bound with the extra J -term; the difference is that Theorem U removes the offset after every mark, while the argument there carries it to the end.

Body-dependent noise. When $\hat{D}_j$ or $\hat{X}_j$ involves the body, their spreads involve the body's state, and the bound must hold for every initial state, including the most benign. A kick $\hat{D} = -\lambda\hat{P}_A + \kappa\hat{y}$ with a fresh probe uncorrelated with the body has $\Delta(\hat{D})^2 = \lambda^2\Delta\hat{P}_A^2 + \kappa^2\Delta\hat{y}^2 \geq \lambda^2\Delta\hat{P}_A^2$, so additive body dependence with a fresh probe adds to the cost. Other forms can lower it for particular states: a kick $-\lambda\hat{P}_A f(\hat{y})$ with f small where some initial state is concentrated costs little on that state, and the theorem then says that the protocol fails for that state unless other marks pay. An apparatus already correlated with the body, which an earlier mark can create, is accounted for through the conditional states.

Why the error drops out. Ozawa's relation bounds a product of error and disturbance and includes the body's own spreads. Theorem U needs only the second factor. The test can separate the hypotheses only if some mark responds differently to the offset, and how a mark responds to a translation of the body is fixed by the generator $U^\dagger G_w U - G_w$, which is a combination of disturbances. The reading error enters the sufficiency side, how close a given protocol comes to the bound.

4. The floors, now complete

Statement	Holds for	Pays in
Phase $\mathcal{K}_\tau/\hbar$ between the motion and its inscribed polygon (polygon-lift note)	Every body state	Nothing: a c-number
$JL + sP \geq \hbar \arcsin(1 - 2\epsilon)$ (probabilistic note)	Every instrument, bounded laboratory	The body's spreads L, P
$\frac{s}{8} \sum \Delta(\hat{D}_j) + \frac{J}{2} \sum \Delta(\hat{X}_j) \geq \hbar \arcsin(1 - 2\epsilon)$ (Theorem U)	Every instrument, every pair of initial states; spreads over the proof's interpolating states	The marks' disturbances
$s \sum \Delta_j \geq 8\hbar \arcsin(1 - 2\epsilon)$ (recoil note)	Body-independent noise, every probe state	Recoil
$\tau \Delta E \geq 24z^2 \hbar \sqrt{(1 - \rho)/(1 + \rho)}$, sharp (mark-cost note)	Gaussian probes with correlation at most ρ	Resolution times recoil

The two universal statements are the second and third rows. Both are hybrid accountings of one offset: the aperture theorem removes it at the body between marks and pays in the

body's spreads; Theorem U carries it through the marks and pays in their disturbances. Both are necessary, and in both Newton's two quantities appear with the same partners: the sagitta with momentum, the impulse with position.

M3, final form. The premise that carries $h > 0$ into the recorded Galileo comparison, for every instrument, is that a record costs disturbance: a mark that registers the sagitta must leave the body's impulse undetermined, and a mark that registers the impulse must leave its position undetermined, with total exchange rate $\hbar$.

5. Consequence for STATE

Theorem A's extension is finished: option 3 (every probe state, the recoil note), option 1 (body-independent noise, the additive-noise note) and option 2 (every instrument, Theorem U). The universal floor for arbitrary instruments is a disturbance bound paired with Newton's sagitta and impulse, and it needs no error–disturbance relation. What remains on the modern leg is to state the combined accounting, which removes the offset at the body in some intervals and carries it through the marks in others, as one optimization, and to decide whether its minimum is attained.